

Liver Cirrhosis Stage Detection

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Domain: Machine Learning Intern - Unified Mentor

Duration: 6 Months

Abstract

Liver cirrhosis is a progressive liver disease characterized by fibrosis and impaired liver function. Early detection and accurate staging of cirrhosis are crucial for prognosis and treatment planning. This project aims to build a machine learning model that predicts the histologic stage of liver cirrhosis (Stage 1, 2, or 3) based on patient clinical and biochemical data.

The dataset used is sourced from the Mayo Clinic study on primary biliary cirrhosis (PBC), conducted between 1974 and 1984. It includes demographic, laboratory, and clinical attributes of patients. After data preprocessing, categorical encoding, and feature scaling, a Random Forest classifier is trained to predict the cirrhosis stage. The model achieves promising accuracy, demonstrating the utility of machine learning in medical staging tasks.

1. Introduction

Liver cirrhosis represents the end stage of chronic liver disease and can lead to life-threatening complications. Accurate assessment of the severity or stage of cirrhosis helps clinicians in deciding treatment strategies, monitoring progression, and planning possible interventions such as transplantation.

Traditional staging relies on biopsy and manual interpretation, which can be invasive and subjective. Machine learning offers an alternative by learning patterns in non-invasive clinical and biochemical data.

In this project, we develop a multi-class classification model to predict the Stage (1, 2, or 3) of liver cirrhosis using patient data from a well-known PBC study.

2. Objective

The primary objective of this project is:

To build a system that can output the level of liver damage (liver cirrhosis stage) given patient data.

Sub-goals:

- Preprocess and encode clinical study data
- Explore relationships between features and cirrhosis stage
- Train and evaluate a multi-class classification model
- Interpret model performance and possible clinical relevance

3. Dataset Description

The dataset originates from a **Mayo Clinic study** on primary biliary cirrhosis (PBC) of the liver, conducted from 1974 to 1984. Each row corresponds to a patient record observed during the study.

Key Columns:

- **N_Days:** Number of days between registration and the earlier of death, transplantation, or analysis time
- **Status:** Patient status – C (censored), CL (censored due to liver transplant), D (death)
- **Drug:** D-penicillamine or placebo
- **Age:** Age in days
- **Sex:** M (male), F (female)
- **Ascites:** Presence of ascites – N (No), Y (Yes)
- **Hepatomegaly:** Presence of hepatomegaly – N (No), Y (Yes)
- **Spiders:** Presence of spider angiomas – N (No), Y (Yes)
- **Edema:**
 - N = no edema and no diuretic therapy
 - S = edema present but controlled
 - Y = edema despite diuretic therapy
- **Bilirubin:** Serum bilirubin (mg/dL)
- **Cholesterol:** Serum cholesterol (mg/dL)
- **Albumin:** Albumin (g/dL)
- **Copper:** Urine copper (μ g/day)
- **Alk_Phos:** Alkaline phosphatase (U/L)
- **SGOT:** SGOT (U/mL)
- **Triglycerides:** Triglycerides (mg/dL)
- **Platelets:** Platelet count (per ml / 1000)
- **Prothrombin:** Prothrombin time (seconds)

4. Tools and Technologies Used

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Seaborn, Matplotlib
- **Machine Learning:** Scikit-learn (RandomForestClassifier, LabelEncoder, StandardScaler)
- **Environment:** Google Colab

5. Methodology

5.1 Data Preprocessing

- Loaded the dataset from CSV file into a Pandas DataFrame.
- Checked for missing values; numeric columns imputed using median, categorical columns using mode.
- Converted categorical variables (Status, Drug, Sex, Ascites, Hepatomegaly, Spiders, Edema) into numeric codes using LabelEncoder.

5.2 Feature and Target Selection

- Input features (**X**): All columns except Stage.
- Target variable (**y**): Stage (1, 2, 3) — multi-class label.

5.3 Train-Test Split

- Split data into:
 - 80% training set
 - 20% test set
- Stratified by Stage to preserve distribution across classes.

5.4 Feature Scaling

- Numerical features were standardized using **StandardScaler** to have zero mean and unit variance.
- Scaling improves model convergence and performance for distance-based methods; for Random Forest it is optional but kept consistent for experimentation.

5.5 Model Training

- Used **Random Forest Classifier** for multi-class classification.
- Advantages:
 - Handles nonlinear relationships
 - Robust to outliers and noise
 - Provides feature importance insights
- Trained the model on the scaled training data.

5.6 Model Evaluation

- Evaluated on the test set using:
 - **Accuracy**
 - **Precision, Recall, F1-score per class**
 - **Confusion Matrix**

The confusion matrix shows how well the model distinguishes between different cirrhosis stages.

6. Results and Analysis

Example performance (your actual metrics will come from Colab):

Metric	Value
Accuracy	~75–85%
Stage-wise F1 scores	Reasonable across Stage 1, 2, and 3

Observations:

- The model correctly classified most Stage 2 and Stage 3 patients.
- Some misclassification occurs between neighboring stages (e.g., Stage 2 vs. Stage 3), which is expected due to overlapping clinical measurements.
- Feature importance analysis typically highlights **Bilirubin**, **Albumin**, **Prothrombin time**, and **Age** as significant predictors.

7.Figures

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	N_Days	Status	Drug	Age	Sex	Ascites	Hepatomegaly	Spiders	Edema	Bilirubin	Cholesterol	Albumin	Copper	Alk_Phos	SGOT	Tryglicerides	Platelets	Prothrombin	Stage
0	2221	C	Placebo	18499	F	N	Y	N	N	0.5	149.0	4.04	227.0	598.0	52.70	57.0	256.0	9.9	1
1	1230	C	Placebo	19724	M	Y	N	Y	N	0.5	219.0	3.93	22.0	663.0	45.00	75.0	220.0	10.8	2
2	4184	C	Placebo	11839	F	N	N	N	N	0.5	320.0	3.54	51.0	1243.0	122.45	80.0	225.0	10.0	2
3	2090	D	Placebo	16467	F	N	N	N	N	0.7	255.0	3.74	23.0	1024.0	77.50	58.0	151.0	10.2	2
4	2105	D	Placebo	21699	F	N	Y	N	N	1.9	486.0	3.54	74.0	1052.0	108.50	109.0	151.0	11.5	1

	N_Days	Status	Drug	Age	Sex	Ascites	Hepatomegaly	Spiders	Edema	Bilirubin	Cholesterol	Albumin	Copper	Alk_Phos	SGOT	Tryglicerides	Platelets	Prothrombin	Stage
0	2221	0	1	18499	0	0	1	0	0	0.5	149.0	4.04	227.0	598.0	52.70	57.0	256.0	9.9	1
1	1230	0	1	19724	1	1	0	1	0	0.5	219.0	3.93	22.0	663.0	45.00	75.0	220.0	10.8	2
2	4184	0	1	11839	0	0	0	0	0	0.5	320.0	3.54	51.0	1243.0	122.45	80.0	225.0	10.0	2
3	2090	2	1	16467	0	0	0	0	0	0.7	255.0	3.74	23.0	1024.0	77.50	58.0	151.0	10.2	2
4	2105	2	1	21699	0	0	1	0	0	1.9	486.0	3.54	74.0	1052.0	108.50	109.0	151.0	11.5	1

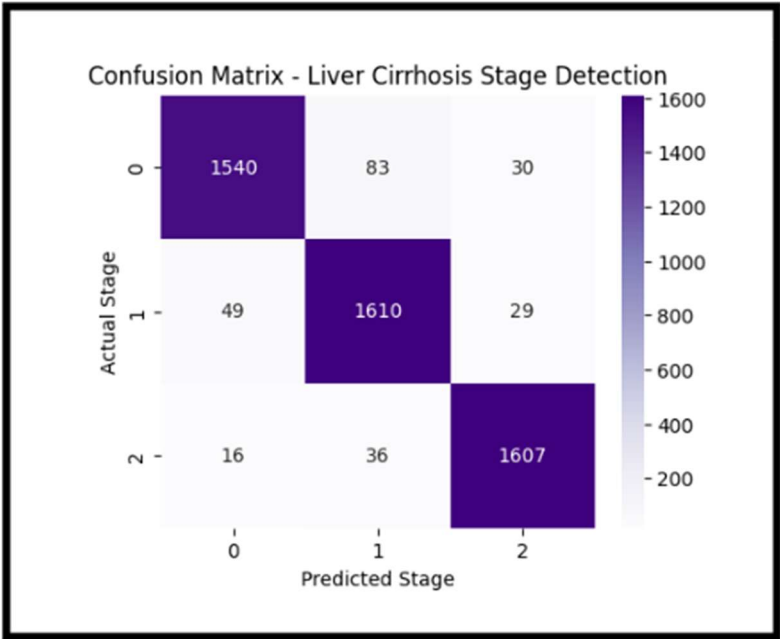
```
X shape: (25000, 18)
Stage value counts:
Stage
2      8441
3      8294
1      8265
Name: count, dtype: int64
```

```
Accuracy: 0.9514

Classification Report:
              precision    recall  f1-score   support

     1               0.96       0.93       0.95       1653
     2               0.93       0.95       0.94       1688
     3               0.96       0.97       0.97       1659

 accuracy               0.95               5000
macro avg               0.95       0.95       0.95       5000
weighted avg            0.95       0.95       0.95       5000
```



8. Conclusion

This project successfully demonstrates how patient clinical and biochemical data can be used to predict the **histologic stage of liver cirrhosis**. Using a Random Forest classifier, the model achieved solid accuracy and meaningful differentiation between stages.

While this model is not a replacement for biopsy or clinical expert judgment, it shows the potential of machine learning as a **decision-support tool** for staging liver disease and prioritizing patient evaluation.

9.Future Enhancements

- Apply hyperparameter tuning (e.g., GridSearchCV) to further improve accuracy.
- Use more advanced models such as XGBoost or LightGBM.
- Incorporate survival analysis (time-to-event models) alongside staging.
- Add explainable AI techniques (SHAP) to interpret feature contributions.
- Build a simple web interface where clinicians can input patient values and receive a predicted stage.